

Application Of Game Theory

What are duopoly games

A **duopoly** is a market structure dominated by **only two firms** that sell a similar or identical product. In Game Theory, we analyze a duopoly not just as a simple market, but as a strategic game where the outcome for each firm depends not only on its own decisions but also on the decisions of its only competitor.

Key Characteristics of a Duopoly

1. **Only Two Sellers:** The entire market supply of a product is controlled by just two firms (Firm A and Firm B).
2. **Interdependence:** This is the most crucial aspect. Firm A's profits are a function of its own actions *and* the actions of Firm B. Firm A cannot make a decision in a vacuum
3. **Strategic Behavior:** Each firm must anticipate the rival's reaction to its own strategies and plan accordingly.

Real World relevance

Coca-Cola & Pepsi: Simple Duopoly Example

- **Market Power:** Together, Coca-Cola (about 50% global share) and Pepsi (about 20%) control ~70% of the world's soft drink market as of 2024.
- **Similar Pricing:** If Coca-Cola sets a price of ₹40 for a 500ml bottle, Pepsi usually sets the same or just below (e.g., ₹39) to stay competitive.
- **Reaction to Price Change:**
 - If **Pepsi drops price** to ₹36, more customers buy Pepsi, so Coca-Cola
 - If **Coca-Cola raises price** to ₹45, many buyers will switch to Pepsi unless Coca-Cola offers something special.
- **Why Not Drop Prices Always?**

- If both try to undercut (price wars), profits for both drop. Typically, they keep prices close to maximize earnings.

Maths used in Game theory (Duopoly)

2. Payoff Matrix

- A **payoff matrix** is a table used in game theory to represent all possible outcomes and payoffs for each player in a strategic situation
- The **rows** list the options or strategies available to one player, while the **columns** list the options for the other player

3. Dominant Strategy

Dominant Strategy → I'm doing the best i can, no matter what you do.

		Firm B	
		Advertise	Dont. Adv.
Firm A	Advertise	10, 5	15, 0
	Dont Adv.	6, 8	10, 2

→ (10, 5)
both advertise no matter what competitors do

4. Nash Equilibrium

Nash Equilibrium → I'm doing the best i can given what you are doing

eg -

		firm B	
		Advertise	Don't Adv.
firm A	Advertise	10, 5	15, 0
	Don't Adv.	6, 8	10, 2

for firm A -

Adv.	10	15
Don't Adv.	6	10

∴ Dominance strategy for A to choose Advertise no matter B chooses what

for firm B -

Adv.	5	8
Don't Adv.	0	2

∴ Dominance strategy for B is only to choose advertise for best he can do.

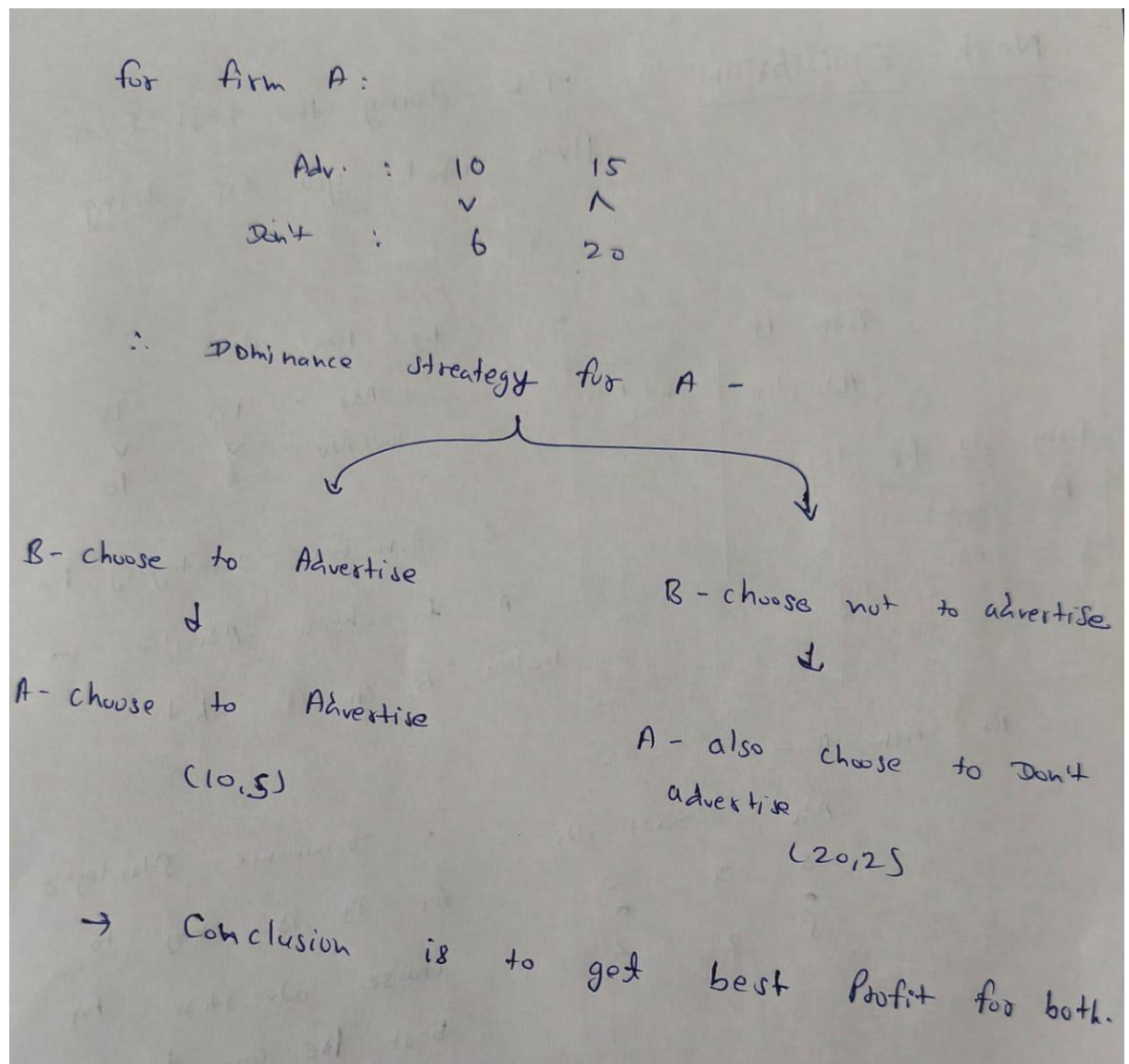
eg:

		firm B	
		Advertise	Don't Adv.
firm A	Advertise	10, 5	15, 0
	Don't Adv.	6, 8	20, 2

for firm B:

Adv.	5	8
Don't	0	2

∴ Dominance strategy for B to choose Advertise not matter whatever A choose



Mathematical Models of Duopoly

1. Cournot Model (Quantity Competition)

- The **Cournot model** describes how two firms compete by choosing how much of a product to produce, not directly by setting prices.
- Each firm's profit depends on **both firms' production levels** because the market drops as total quantity rises.
- Firms make their output decisions **simultaneously and independently**, each assuming the other's production is fixed. This creates a cycle of reaction and adjustment that eventually settles at a stable outcome, called the **Cournot equilibrium**

- The model shows that in equilibrium, **firms split the market and earn moderate profits**—less than if they acted together like a monopoly, but more than in a perfectly competitive market.

Cournot Model

↳ Quantity Competition

for Duopoly -

two companies in competition

$q_1, q_2 \rightarrow$ Quantities produced by firm 1 and firm 2

Total Quantity : $Q = q_1 + q_2$

Market Price : $P(Q) = a - bQ$

$a, b \rightarrow$ demand Parameters

marginal cost = c

Profit fn. for firm 1:

$$\pi_1 = (a - b(q_1 + q_2))q_1 - cq_1$$

for profit-maximize -

$$\frac{d\pi_1}{dq_1} = a - 2bq_1 - bq_2 - c = 0$$

$$\Rightarrow q_1^* = \frac{a - c - bq_2}{2b} \quad \text{--- (I)}$$

Profit fn. for firm 2

$$\pi_2 = (a - b(q_1 + q_2))q_2 - cq_2$$

for profit-maximize -

$$\frac{d\pi_2}{dq_2} = a - bq_1 - 2bq_2 - c = 0$$

$$\Rightarrow q_2^* = \frac{a - c - bq_1}{2b} \quad \text{--- (II)}$$

from $EQ^1(u)$ & $EQ^2(u)$ -

$$q_1^* = \frac{a - c - b \left(\frac{a - c - b q_1^*}{2b} \right)}{2b} \Rightarrow q_1^* = \frac{a - c}{3b}$$

$$q_2^* = \frac{a - c - b \left(\frac{a - c - b q_2^*}{2b} \right)}{2b} \Rightarrow q_2^* = \frac{a - c}{3b}$$

$$q_1^* = q_2^* = \frac{a - c}{3b}$$

↳ Cannot equilibrium point

for different marginal cost -

$$q_1^* = \frac{a - c_1 - b q_2}{2b} \rightarrow q_1^* = \frac{a - 2c_1 + c_2}{3b}$$

$$q_2^* = \frac{a - c_2 - b q_1}{2b} \rightarrow q_2^* = \frac{a - 2c_2 + c_1}{3b}$$

→ Firm 1's best response $\Rightarrow$ when $q_2 = 0$

→ Firm 2's best response $\Rightarrow$ when $q_1 = 0$

from $\pi_1^*(q_2)$ & $\pi_2^*(q_1)$ -

$$q_1^* = \frac{a - c - b \left(\frac{a - c - b q_1^*}{2b} \right)}{2b} \Rightarrow q_1^* = \frac{a - c}{3b}$$

$$q_2^* = \frac{a - c - b \left(\frac{a - c - b q_2^*}{2b} \right)}{2b} \Rightarrow q_2^* = \frac{a - c}{3b}$$

$$q_1^* = q_2^* = \frac{a - c}{3b}$$

↳ Cournot equilibrium point

for different marginal cost -

$$q_1^* = \frac{a - c_1 - b q_2}{2b} \rightarrow q_1^* = \frac{a - 2c_1 + c_2}{3b}$$

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→ firm 1's best response $\Rightarrow$ when $q_2 = 0$

→ firm 2's best response $\Rightarrow$ when $q_1 = 0$

2. Bertrand Model (Price Competition)

1. Theory and Explanation

- The **Bertrand model** describes competition between two firms that sell identical (homogeneous) products and compete solely by choosing their selling prices, **not quantities**.
- Consumers always buy from the firm offering the **lower price**, as products are seen as perfect substitutes. If both firms charge the same price, they split the market equally.
- Firms choose prices **simultaneously**, each aiming to maximize profit, while assuming the other's price is fixed.
- If either firm tries to set a price just a bit lower than the rival, it captures the whole market. This leads to an intense price-cutting competition, sometimes called a "race to the bottom."
- The Bertrand model predicts that this process continues until **both firms' prices equal marginal cost** (unit cost of production). At this point, neither firm earns economic profit and any attempt to raise price means losing all customers to the rival.

- **Mathematical Model**

Let both firms have **marginal cost** c (for example, $c=10$).

Each firm sets price: p_1 (Firm 1), p_2 (Firm 2). **Firm**

Profits

- If $p_1 < p_2$: Firm 1 sells to the whole market, Firm 2 sells nothing.
- If $p_1 > p_2$: Firm 2 sells to the whole market, Firm 1 sells nothing.
- If $p_1 = p_2$: Both split sales equally.

Profit for a firm:

$$\begin{aligned}\pi_1 &= (p_1 - c) \cdot D(p_1) \text{ if } p_1 < p_2 \\ \pi_1 &= 0 \text{ if } p_1 > p_2 \\ \pi_1 &= 0.5 \cdot (p_1 - c) \cdot D(p_1) \text{ if } p_1 = p_2\end{aligned}$$

Where $D(p_1)$ is the total demand at price p_1 .

Nash Equilibrium

- The only stable equilibrium is where $p_1 = p_2 = c$.
- No firm can raise price without losing all market share, and lowering price below cost is unprofitable.